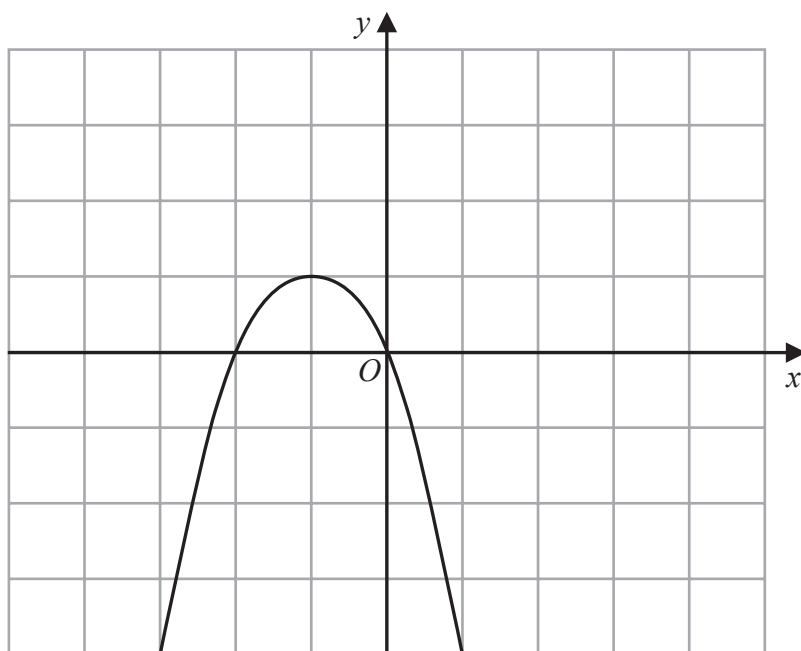


21 The graph of $y = f(x)$ is shown on the grid below.



(a) On the grid above, sketch the graph of $y = -f(x)$

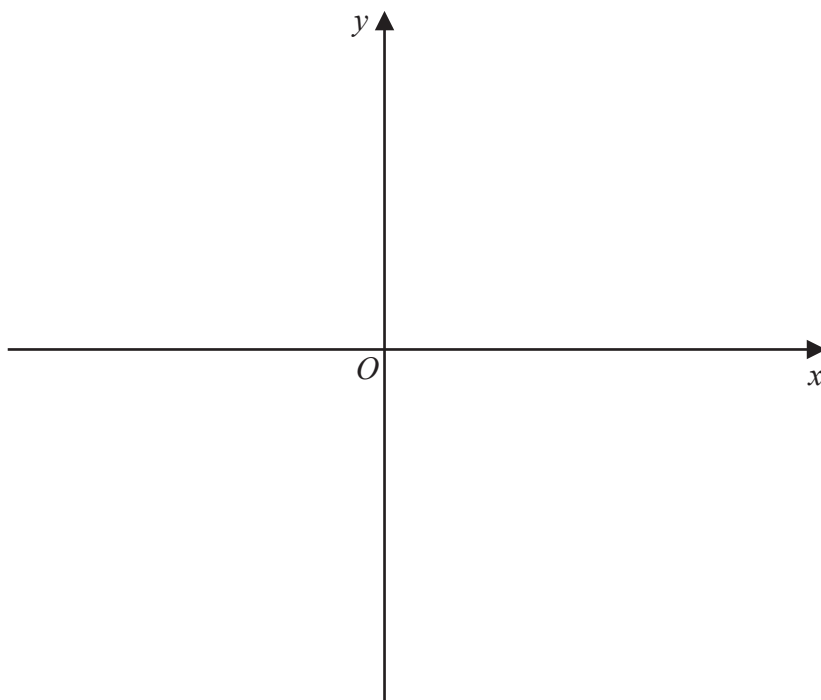
(1)

C is the circle with equation $x^2 + y^2 = 9$

The circle **C** is translated by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$ to give circle **S**.

(b) Draw a sketch of circle **S**.

Label with coordinates the centre of circle **S** and any points of intersection with the x -axis.



(3)

T is a circle with centre $(0, 0)$

P is the point on **T** with coordinates $(12, -9)$

(c) Find an equation of the tangent to **T** at the point P .

.....
(3)

(Total for Question 21 is 7 marks)